



Install Required Libraries

```
In [1]: pip install gymnasium numpy matplotlib
```

```

Collecting gymnasium
  Downloading gymnasium-1.3.0-py3-none-any.whl.metadata (10 kB)
WARNING: Cache entry deserialization failed, entry ignored
Collecting numpy
  Using cached numpy-2.2.6-cp310-cp310-macosx_14_0_arm64.whl.metadata (62 kB)
WARNING: Cache entry deserialization failed, entry ignored
Collecting matplotlib
  Downloading matplotlib-3.10.9-cp310-cp310-macosx_11_0_arm64.whl.metadata (52
kB)
Collecting cloudpickle>=1.2.0 (from gymnasium)
  Using cached cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: typing-extensions>=4.3.0 in /Users/mohammadadnan
ali/Library/Python/3.10/lib/python/site-packages (from gymnasium) (4.15.0)
Collecting farama-notifications>=0.0.1 (from gymnasium)
  Downloading farama_notifications-0.0.6-py3-none-any.whl.metadata (729 bytes)
WARNING: Cache entry deserialization failed, entry ignored
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.2-cp310-cp310-macosx_11_0_arm64.whl.metadata (5.5 k
B)
WARNING: Cache entry deserialization failed, entry ignored
Collecting cycler>=0.10 (from matplotlib)
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
WARNING: Cache entry deserialization failed, entry ignored
Collecting fonttools>=4.22.0 (from matplotlib)
  Downloading fonttools-4.63.0-cp310-cp310-macosx_10_9_universal2.whl.metadata
(118 kB)
WARNING: Cache entry deserialization failed, entry ignored
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.0-cp310-cp310-macosx_11_0_arm64.whl.metadata (5.1
kB)
Requirement already satisfied: packaging>=20.0 in /Users/mohammadadnanali/Libra
ry/Python/3.10/lib/python/site-packages (from matplotlib) (26.2)
WARNING: Cache entry deserialization failed, entry ignored
Collecting pillow>=8 (from matplotlib)
  Downloading pillow-12.3.0-cp310-cp310-macosx_11_0_arm64.whl.metadata (9.1 kB)
WARNING: Cache entry deserialization failed, entry ignored
Collecting pyparsing>=3 (from matplotlib)
  Using cached pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: python-dateutil>=2.7 in /Users/mohammadadnanali/
Library/Python/3.10/lib/python/site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /Users/mohammadadnanali/Library/Pyth
on/3.10/lib/python/site-packages (from python-dateutil>=2.7->matplotlib) (1.1
7.0)
Downloading gymnasium-1.3.0-py3-none-any.whl (953 kB)
_____ 953.9/953.9 kB 2.6 MB/s 0:00:0036
m-:--:--
Using cached numpy-2.2.6-cp310-cp310-macosx_14_0_arm64.whl (5.3 MB)
Downloading matplotlib-3.10.9-cp310-cp310-macosx_11_0_arm64.whl (8.2 MB)
_____ 8.2/8.2 MB 4.7 MB/s 0:00:01 eta
0:00:01
Using cached cloudpickle-3.1.2-py3-none-any.whl (22 kB)
Downloading contourpy-1.3.2-cp310-cp310-macosx_11_0_arm64.whl (253 kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading farama_notifications-0.0.6-py3-none-any.whl (2.9 kB)

```

```
Downloading fonttools-4.63.0-cp310-cp310-macosx_10_9_universal2.whl (2.9 MB)
_____ 2.9/2.9 MB 4.4 MB/s 0:00:00m0:00:0
100:01
Downloading kiwisolver-1.5.0-cp310-cp310-macosx_11_0_arm64.whl (63 kB)
Downloading pillow-12.3.0-cp310-cp310-macosx_11_0_arm64.whl (4.8 MB)
_____ 4.8/4.8 MB 6.8 MB/s 0:00:00 eta
0:00:01
Using cached pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: farama-notifications, pyparsing, pillow, numpy,
kiwisolver, fonttools, cyclor, cloudpickle, gymnasium, contourpy, matplotlib
_____ 11/11 [matplotlib]1 [matplotlib]
Successfully installed cloudpickle-3.1.2 contourpy-1.3.2 cyclor-0.12.1 farama-n
otifications-0.0.6 fonttools-4.63.0 gymnasium-1.3.0 kiwisolver-1.5.0 matplotli
b-3.10.9 numpy-2.2.6 pillow-12.3.0 pyparsing-3.3.2

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: /opt/homebrew/opt/python@3.10/bin/python3.10 -m pip in
stall --upgrade pip
Note: you may need to restart the kernel to use updated packages.
```

Import Libraries

```
In [2]: import gymnasium as gym
import numpy as np
import matplotlib.pyplot as plt
```

Task 1: Verify Installation

```
In [3]: print("Gymnasium Version:", gym.__version__)
```

Gymnasium Version: 1.3.0

Task 2: Create and Initialize CartPole Environment

```
In [4]: # Create CartPole environment
env = gym.make("CartPole-v1")

# Reset environment
observation, info = env.reset()

# Display initial observation and environment info
print("Initial Observation:")
print(observation)

print("\nEnvironment Info:")
print(info)
```

Initial Observation:

```
[ 0.04493817 -0.01148041  0.03236548 -0.03369409]
```

Environment Info:

```
{}
```

Task 3: Explore Observation and Action Spaces

```
In [5]: print("Observation Space:")
        print(env.observation_space)

        print("\nAction Space:")
        print(env.action_space)

        print("\nType of Observation Space:")
        print(type(env.observation_space))

        print("\nNumber of Possible Actions:")
        print(env.action_space.n)
```

Observation Space:

```
Box([-4.8          -inf -0.41887903          -inf], [4.8          inf
0.41887903          inf], (4,), float32)
```

Action Space:

```
Discrete(2)
```

Type of Observation Space:

```
<class 'gymnasium.spaces.box.Box'>
```

Number of Possible Actions:

```
2
```

Task 4: Execute a Random Agent

```
In [6]: # Reset environment
        observation, info = env.reset()

        done = False
        step = 0
        total_reward = 0

        print("-" * 70)

        while not done:

            # Select random action
            action = env.action_space.sample()

            # Execute action
            observation, reward, terminated, truncated, info = env.step(action)
```

```
done = terminated or truncated

step += 1
total_reward += reward

print(f"Step {step}")
print(f"Action: {action}")
print(f"Observation: {observation}")
print(f"Reward: {reward}")
print(f"Episode Finished: {done}")
print("-" * 70)

print("Episode Completed!")
print("Total Steps:", step)
print("Cumulative Reward:", total_reward)

env.close()
```

Step 1
Action: 0
Observation: [0.03080072 -0.21804488 -0.01671047 0.31777993]
Reward: 1.0
Episode Finished: False

Step 2
Action: 0
Observation: [0.02643982 -0.4129249 -0.01035487 0.60514647]
Reward: 1.0
Episode Finished: False

Step 3
Action: 0
Observation: [0.01818132 -0.6079005 0.00174806 0.89454997]
Reward: 1.0
Episode Finished: False

Step 4
Action: 0
Observation: [0.00602331 -0.8030461 0.01963906 1.1877818]
Reward: 1.0
Episode Finished: False

Step 5
Action: 1
Observation: [-0.01003761 -0.6081842 0.0433947 0.9013189]
Reward: 1.0
Episode Finished: False

Step 6
Action: 0
Observation: [-0.0222013 -0.8038664 0.06142107 1.2073199]
Reward: 1.0
Episode Finished: False

Step 7
Action: 0
Observation: [-0.03827862 -0.9997257 0.08556747 1.5186013]
Reward: 1.0
Episode Finished: False

Step 8
Action: 1
Observation: [-0.05827314 -0.80573636 0.1159395 1.253808]
Reward: 1.0
Episode Finished: False

Step 9
Action: 1
Observation: [-0.07438787 -0.6122743 0.14101566 0.999573]
Reward: 1.0
Episode Finished: False

Step 10
Action: 1
Observation: [-0.08663335 -0.41928995 0.16100712 0.75429124]
Reward: 1.0
Episode Finished: False

Step 11
Action: 1
Observation: [-0.09501915 -0.22671053 0.17609294 0.5162901]
Reward: 1.0
Episode Finished: False

Step 12
Action: 0
Observation: [-0.09955336 -0.42381784 0.18641874 0.8588831]
Reward: 1.0
Episode Finished: False

Step 13
Action: 1
Observation: [-0.10802972 -0.23165752 0.20359641 0.6301268]
Reward: 1.0
Episode Finished: False

Step 14
Action: 1
Observation: [-0.11266287 -0.03987079 0.21619894 0.40783316]
Reward: 1.0
Episode Finished: True

Episode Completed!
Total Steps: 14
Cumulative Reward: 14.0